OBJECT ORIENTED PROGRAMMING USING VB.NET

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Recommended Textbooks

1. **Mike Snell & Lars Powers**; "Microsoft Visual Studio 2010 Unleash" Pearson Education, Inc.
2. **Evangelos Petroutsos**, "Mastering Microsoft Visual Basic 2010", Wiley Publishing Inc.

MATHEMATICAL OPERATORS

- An **operator** in a programming language is a symbol that tells the compiler or interpreter to perform specific mathematical, relational or logical operations and give out a result. Operators are widely used in computer programming to perform computations and calculations.

- Some operators used in programming includes;
- **Arithmetic operators**
- **Comparison / Relational operators**
- **Logical operators**
- **Compound operators**
- **Increment operators**
- **Decrement operators**
- **Unary operators**

MATHEMATICAL OPERATORS

- **Arithmetic operators:** Arithmetic operations are performed according to the rules of BODMAS in mathematics.

- The basic arithmetic operations includes;
- Addition (+): Use to find sum
- Subtraction (-): Use to find difference
- Multiplication (*): Use to find product
- Division (/): Use to find division
- Modulo (%): Use to find remainder

- The Modulo (%) is a special operator that is used to find the remainder when a number divides another number.

MATHEMATICAL OPERATORS

- **Comparison / Relational operators:** The comparison or relational operators are used to compare two or more values in programming.

- The comparison operators includes;
- Less Than (<)
- Greater Than (>)
- Less or Equal to (<=)
- Greater or Equal to (>=)
- Equal to (==)
- Not Equal to (!=)

MATHEMATICAL OPERATORS

- **Logical operators:** The logical operators are used to perform logic and Boolean operations in programming.

- The logical operators includes;
- AND (&&)
- OR (||)
- NOT Gate (!)
- NAND Gate ()
- NOR ()
- XOR ()
- XNOR ()

MATHEMATICAL OPERATORS

- **Compound operators:** The compound operators are a composition of arithmetic operators used to perform special functions in programming.

- Some compound operators includes;
- **Add and Assign (+=)**: Used to add values and the result is stored back into the same memory
- **Subtract and Assign (-=)**: Used to subtract values and the result is stored back into the same memory
- **Multiply and Assign (*=)**: Used to multiply values and the result is stored back into the same memory
- **Divide and Assign (/=)**: Used to divide values and the result is stored back into the same memory
- **Example**: X = X + 5; This can be written as X+=5;

  Y = Y – 10; This can be written as Y-=10;

MATHEMATICAL OPERATORS

- **Increment operator:** The increment operator is used to add one to a value in memory.
- Example:
- X = X +1; This can be written as X+=1;
- Y = Y+1; This can be written as Y+=1;

- **Decrement operator:** The increment operator is used to subtract one from a value in memory.
- Example:
- X = X -1; This can be written as X-=1;
- Y = Y-1; This can be written as Y-=1;

MATHEMATICAL OPERATORS

- **Unary operators:** The unary operator is used to perform operations on a single operand or variable.

- Some examples of unary operators includes;
- **Plus (+)**: The unary plus operator precedes its operand and converts it into a positive number. E.g. +5, +12

- **Minus (-)**: The unary minus operator precedes its operand and converts it into a negative number. E.g. -5

- **Ampersand(&)**: The unary ampersand operator precedes its operand and is used to get the address location of the variable. E.g. &X, &Y, &Name, &Age

THE END

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